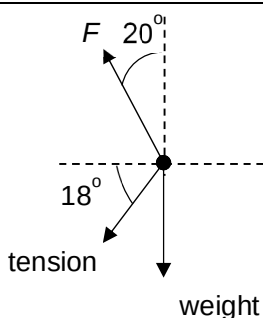


	$V_{ft} = 1.29 V$	fraction)
3(a)	Since aeroplane is <u>constantly changing direction</u>, and hence velocity, it <u>must have a force</u> acting on it by <u>Newton's First Law</u>. Since aeroplane is <u>moving in a circle</u>, the <u>change in velocity</u> at every instant is <u>directed towards the centre of its path</u>. By <u>Newton's Second Law</u>, the force must hence be <u>directed towards the centre of its path</u> too.	B1 B1
3(b)	 B2: all 3 forces with correct angles B1: missing 1 force / unclear labels / unclear angles / drew in centripetal force 0: 2 or more mistakes committed listed for B1	

Qns	Answer	Marks
3(c)	Vertical : $F \cos(20^\circ) = T \sin(18^\circ) + mg$ $= T \sin(18^\circ) + (0.75)(9.81)$ Horizontal : $F \sin(20^\circ) + T \cos(18^\circ) = F_c = \frac{mv^2}{r} = \frac{(0.75)(35)^2}{60 \cos(18^\circ)}$ Using GC: $T = 12.6 \text{ N}$ $(F = 12.0 \text{ N})$	C1 (vertical equilibrium) C1 (horizontal net force) A1
4(a)	gravitational force per unit mass	B1